VOICE-FIRST LOAN COMPREHENSION VERIFICATION ON GOOGLE GEMMA 4

Ardham

Oral Loan Verification & Empowerment

A signature is proof of consent. Ardham is proof of comprehension.

- 100% Offline-Ready — runs on-device with Wi-Fi switched OFF
  - SYSTEM STATUS: 100%
- Google Gemma 4 — Apache 2.0 open-weight multimodal SLM
  - SLM EFFICIENCY: HIGH
- Lakshmi AI Avatar — real-time procedural voice lip-sync
  - LIP-SYNC PRECISION
- Tamper-Proof Seal — SHA-256 hash plus scannable QR receipt
  - SECURITY VERIFIED

Major League Hacking | Hack Day Vizag × AI for Vizag — Track badges: Build with Gemma • Best Open-Source AI Project



Ardham (అర్ధం) | Hacktoberfest × AI for Vizag | MLH | Track: Build with Gemma & Best Open-Source AI

The Rural Lending Crisis
Signed, But Never Understood



Technical Core

Gemma 4 On-Device Pipeline & Deterministic Guard

1. **Vision Extraction** → Gemma 4 E4B extracts financial terms



2. **Deterministic Guard** → Math check (Installment × Weeks = Total)



3. **Spoken Mother Tongue** → Telugu & Hindi with lip-sync



4. **Tamper-Proof Seal** → SHA-256 certificate + QR audit



100% Offline-Ready (Ollama / Local SLM) | **26/26** Pytest Tests Passing | **<270ms** Frontend Build

Lakshmi AI Avtar

100% Eyes-Free Audio Verification for Blind Borrowers



Lakshmi
AI Loan Officer

Procedural Rig

Procedural Lip-Sync Rig

Real-time 60 FPS, natural voice & gestures

Hands-Free Access

Hands-Free Voice Commander

Say “Next”, “Back”, “Repeat”

Simple & Clear

Side-Docked Ephemeral Cards

Key loan details for 5 seconds

Zero Technical Jargon

Plain language: “Easy weekly repayments”

Instant 1-Click Avatar Test

Smart Experience Flow

Speak Naturally → Get Key Info (5 sec) → Complete Verification

Greater Inclusion. Stronger Financial Access.

100% Hands-Free

Inclusive for Blind & Deaf

Built for Rural India

MAJOR LEAGUE HACKING

Track Alignment

Build with Gemma & Best Open-Source AI

Build with Gemma (Google DeepMind Track)

100% PASS

- Google Gemma 4 open weights
- Edge SLM + Cloud LLM

Best Open-Source AI Project (MLH Track)

100% PASS

- 100% Public GitHub • Apache 2.0

Agent Skills Open Standard Compliance

PASS

- Modular loan products in 'skills/'

Production Cloud & Domain Deployment

PASS

- Deployed on Render • ardham.tech

Team - Ardham

Why Ardham Wins

Authentic Ground Impact
Solves **rural debt traps** before any thumbprint or signature.

Zero-Hallucination Engineering
Deterministic math, not chatbot guesses.

Built for India's Next Billion
100% offline-ready, hands-free for all.



Guru Sai Sumith
CEO – Ardham

Full Stack
AI-ML

Production - Live

